885406
Single-shell in-situ concrete tunnel lining: experience in the Federal Republic of Germany
Haack, A
Tunnlg Underground Space Technol V3, N1, 1988, P55-66

Recent developments in the use of shotcrete and steel fibre reinforced pumping concrete for tunnel linings are presented. In shotcrete construction, the effects of external load, temperature and shrinkage are discussed, along with experiments performed in two test tunnels where magnitude of shrinkage deformation was studied. For pumped concrete, roller ring seal and inspection joint strip tests are presented. More recently, the development and application of a novel tunnel lining for large cross sections using steel plate skin and unreinforced backfill concrete has been undertaken.

885407
TBM tunnelling in poor and very poor rock conditions
Martin, D
Tunn Tunnlg V20, N3, March 1988, P22-28

TBM tunnelling is most economical for long drivages in good rock and problems arise in poor ground. Two case studies of successful drivage in Italy, where zones of poor to very poor rock were encountered, are presented. Support by shotcrete, mesh, and swelling bolts was used to stabilise in front of the cutterhead, and then the machine bored through. Advance ahead of the cutter was by hand mining at worst. Knowledge gained from driving a pilot tunnel proved valuable.

885408
Comparison of tunnelling methods adopted on the Singapore MRT
Hulme, T; Copsey, P
Tunn Tunnlg V20, N3, March 1988, P31-38

The 67km project went through sandstone, siltstone, and granite valleys infilled with estuarine, fluvial, marine, and beach deposits. Rocks varied from intact to fully weathered. The use of a shield with roadheader or backhole excavation was most common, with few TBM stretches. NATM tunnelling was employed in several contracts. Silica fume was used in the shotcrete with advantages in ease of handling and lowered permeability. Final lining was segmented concrete.

885409
Hammer tunnelling comes of age
Smith, M
World Tunnlg V1, N1, March 1988, P32-36

The use of the hydraulic impact hammer in full face tunnelling is a recent development. The method at present is most suited to fissured and layered strata. Advantages of the method include loading out spoil concurrently with the excavation process, high productivity rates and low capital costs. Case studies at Reiti road tunnel and San Remo rail tunnel,both in Italy, illustrate use of the technique.

885410
Channel rail tunnel gets underway
Smith, M
World Tunnlg V1, N1, March 1988, P10-24

A review of progress to date in the construction of the Channel Tunnel is presented. The geology, tunnel linings, tunnel boring machines, ancillary equipment and facilities, spoil handling and disposal, finance, ancillary works and potential problems are discussed.

885411
Bouygues tunnel boring machine placed into operation for main stone-drift mining in the workings of Kempense Steenkolenmijnen
Philippron, P
Proc 13th World Mining Congress, Improvement of Mine Productivity and Overall Economy by Modern Technology, Stockholm, June 1987 V2, P633-641. Publ Rotterdam: A A Balkema, 1987

The Campine Basin in Belgium is subject to high ground pressures exerted by water bearing non-carboniferous overburden. Past experiences of stone-drift mining and erection of segmental reinforced concrete linings are summarised. The use of a TBM in faulty conditions liable to bed separation is described. Initial results after 2116m of driving indicate that major fault zones can be traversed relatively easily and economically using this machinery.

885412
Use of a compact full-face tunnelling machine for in-seam heading on Warndt colliery
Dietrich, W
Proc 13th World Mining Congress, Improvement of Mine Productivity and Overall Economy by Modern Technology, Stockholm, June 1987 V2, P643-652. Publ Rotterdam: A A Balkema, 1987

Classic full face tunnelling machines are uneconomical for short drivages. For use where there are hard abrasive rocks, beyond the capacity of a point attack roadheader, a new compact tunnelling machine has been developed. The machine and its initial trials are described. Results indicate this type of machine could become widely adopted in mechanised headings for development projects from 1500m upwards, or for in seam roadways with hard abrasive adjacent rock.

885413
Displacement behaviour around tunnel intersection in weak rock
Takebasyashi, T; Yasukochi, T; Matsumoto, S
Proc International Symposium on Engineering in Complex Rock Formations, Beijing, 3-7 November 1986 P839-845. Publ Beijing: Science Press, 1986

The New Austrian Tunnelling Method was used in Japan for the first time in the construction of a dust collection tunnel for a roadway tunnel. The ground was layered Palaeozoic strata. The reinforcement used and horizontal convergence and roof settlement observed are discussed. Displacements were relatively large due to the presence of a phyllitic fracture zone.

885414
Stress behaviour and ground pressure of tunnel with double lining
Zhao Zirong
Proc International Symposium on Engineering in Complex Rock Formations, Beijing, 3-7 November 1986 P929-935. Publ Beijing: Science Press, 1986

Measurements from field and model tests of stress distribution and ground pressure around double lined tunnels in soft ground are presented. Results suggest that the surrounding rock can be treated as an integral part of the supporting structure. The bearing capacity of the double lining is greater than that of a single lining of similar thickness. The time dependences of ground pressure and load on the lining are also demonstrated.

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